Zonal Informatics Olympiad, 2025

Solutions

1. Minimum number of operations to obtain a number greater than X.

   (a) $N = 2$, $X = 77$
   Answer: 3

   (b) $N = 10$, $X = 0000942274$
   Answer: 1

   (c) $N = 15$, $X = 127932412749752$
   Answer: 1

2. Number of passcodes you can create of length N

   (a) $N = 3$
   Answer: 570

   (b) $N = 6$
   Answer: 73

   (c) $N = 11$
   Answer: 47

3. Minimum number of operations to make the row of towers special

   (a) $N = 5$, $H = [3, 1, 3, 4, 1]$
   Answer: 2

   (b) $N = 12$, $H = [12, 9, 10, 7, 11, 9, 4, 6, 15, 5, 9, 12]$
   Answer: 34

   (c) $N = 25$, $H = [22, 26, 7, 25, 26, 14, 22, 26, 8, 20, 2, 22, 24, 6, 6, 10, 11, 22, 20, 22, 3, 10, 28, 25, 15]$
   Answer: 262

4. Number of arrays of pairs

   (a) $N = 30$, $P = [-1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29]$
   Answer: 185

   (b) $N = 15$, $P = [-1, 1, 1, 2, 3, 1, 2, 2, 3, 3, 5, 5, 4, 11, 12]$
   Answer: 836

   (c) $N = 30$, $P = [-1, 1, 1, 2, 2, 3, 1, 1, 2, 2, 2, 3, 4, 1, 1, 1, 2, 3, 3, 1, 4, 4, 5, 6, 5, 6, 6, 3, 3, 10]$
   Answer: 451

Marking

The question paper carries 80 marks, broken up into four questions of 20 marks each. Each question has three parts. *If you solve all three parts correctly, you get 20 marks for that question.* Otherwise, you get 5 marks for each part that you solve correctly.

Qualifying cutoff

Male students

- Std 12: 40
- Std 11: 40
- Std 10: 35
- Std 9: 30
- Std 8: 25
- Std 7 and below: 20

Female students

- Std 12: 25
- Std 11: 25
- Std 10: 25
- Std 9: 20
- Std 8: 20
- Std 7 and below: 20

Score distribution

Score	80	75	70	65	60	55	50	45	40	35	30	25	20	15	10	5	0
Number at or above this score	0	0	0	0	0	3	17	27	50	88	145	238	299	325	362	385	421